2 **(a)** State the conditions required for an object to be in equilibrium.

....

....

.... [2]

(b) A block rests in equilibrium on the centre of a uniform horizontal plank. The block is attached to two ropes, each of which passes over a fixed pulley and then attaches to an end of the plank as shown in Figure 2.1.

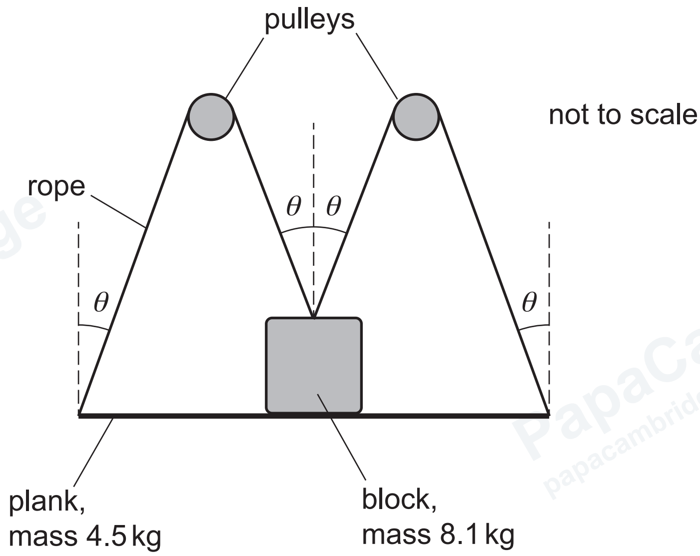

Figure 2.1

Where the ropes attach to the block and to the plank, the angle between the ropes and the vertical is θ. There is negligible friction between the pulleys and the ropes.

The mass of the block is 8.1 kg. The mass of the plank is 4.5 kg.

Figure 2.2 shows the directions of the forces acting on the plank and the block.
The weight of the plank is W_P, the weight of the block is W_B, the tension in each rope is T and the forces that the plank and the block exert on each other each have magnitude R.

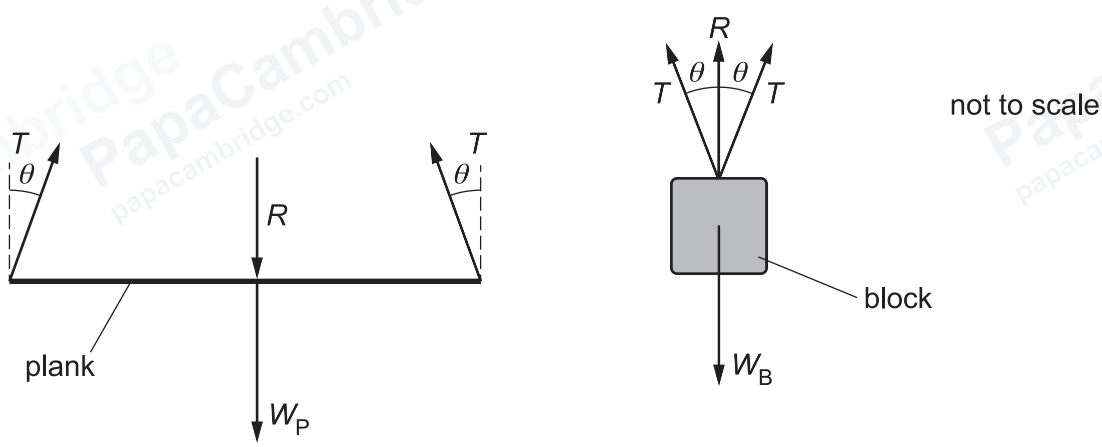

Figure 2.2

(i) Use Figure 2.2 to show that

$$R + W_P = W_B - R.$$

[2]

(ii) Use the equation in **2(b)(i)** to calculate the magnitude of R.

R = ... N [2]

(c) The extension of each rope in **2(b)** is proportional to the force applied to it.

(i) State the name of the law that describes this behaviour.

.... [1]

(ii) The ropes have elastic deformation.

State what is meant by elastic deformation.

....

.... [1]

[Total: 8]